HandsOn-14

Developers of Apps Centric Solutions have created an employee management application which supports light and dark themes for the buttons. The current solution uses the react state and props to provide the theme name to be used from App component to Employee List component and from there to Employee Card component. Quality assurance team analyzed the solutions and found the technique being used to be a substandard one. React architect suggested to use the react context API to share the theme name with nested child components instead of passing them down using props from the parent component.

You are assigned the task of converting the application form props only to React Context API.

.

Implementation

→App.js

```
// src/App.js
import React, { useState } from 'react';
import EmployeesList from './EmployeesList';
import ThemeContext from './ThemeContext';
function App() {
  const [theme, setTheme] = useState('light');

  const toggleTheme = () => {
    setTheme(prev => prev === 'light' ? 'dark' : 'light');
  };
  return (
    <ThemeContext.Provider value={theme}>
      <div style={{ padding: "20px" }}>
        <button onClick={toggleTheme}>
          Toggle Theme (Current: {theme})
        </button>
        <EmployeesList />
      </div>
```

```
    </ThemeContext.Provider>
  );
}
export default App;
```

→App.css

```css
.App {
  text-align: center;
}
.App-logo {
  height: 40vmin;
  pointer-events: none;
}
@media (prefers-reduced-motion: no-preference) {
  .App-logo {
    animation: App-logo-spin infinite 20s linear;
  }
}
.App-header {
  background-color: #282c34;
  min-height: 100vh;
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  font-size: calc(10px + 2vmin);
  color: white;
}
.App-link {
  color: #61dafb;
```

```
}
@keyframes App-logo-spin {
  from {
    transform: rotate(0deg);
  }
  to {
    transform: rotate(360deg);
  }
}
```

→EmployeeCard.js

```
// src/EmployeeCard.js
import React, { useContext } from 'react';
import ThemeContext from './ThemeContext';
function EmployeeCard({ employee }) {
  const theme = useContext(ThemeContext);
  const buttonStyle = {
    backgroundColor: theme === 'light' ? '#eee' : '#333',
    color: theme === 'light' ? '#000' : '#fff',
    border: '1px solid #ccc',
    padding: '10px',
    marginTop: '10px'
  };
  return (
    <div style={{ marginBottom: '15px' }}>
      <h3>{employee.name}</h3>
      <p>{employee.designation}</p>
      <button style={buttonStyle}>Contact</button>
    </div>
  );
```

```
}
export default EmployeeCard;
```

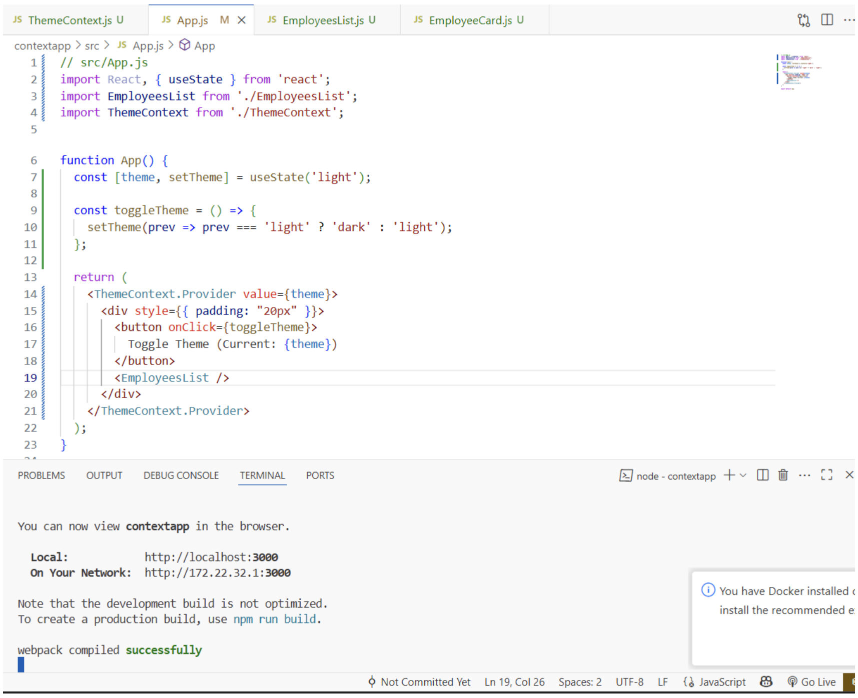
➔EmployeesList.js

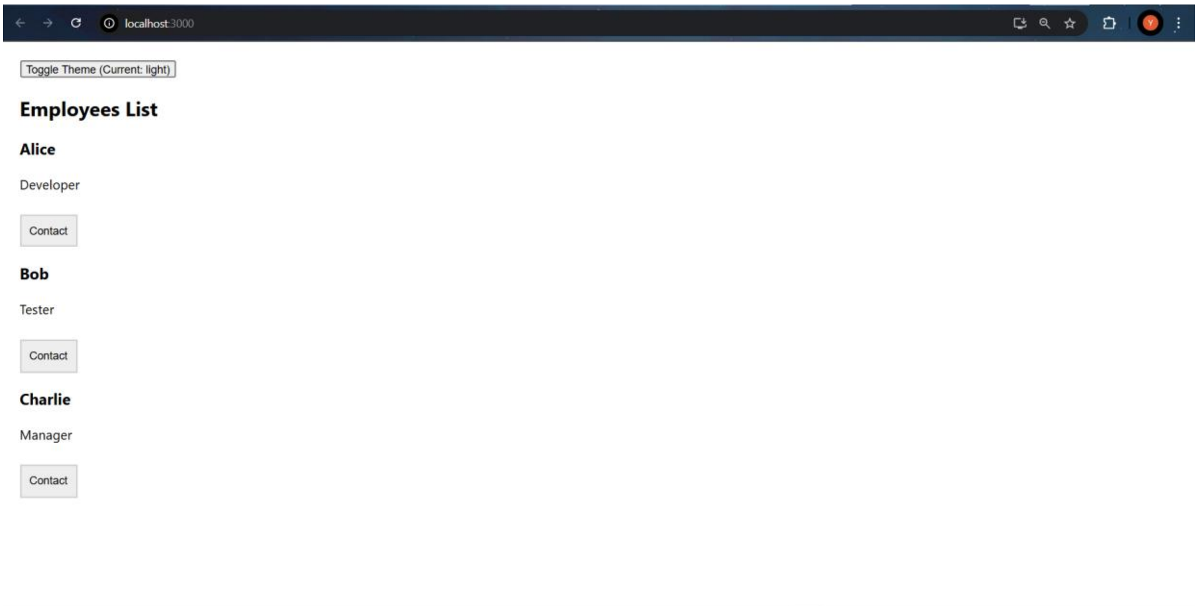
```
// src/EmployeesList.js
import React from 'react';
import EmployeeCard from './EmployeeCard';
function EmployeesList() {
  const employees = [
    { id: 1, name: 'Alice', designation: 'Developer' },
    { id: 2, name: 'Bob', designation: 'Tester' },
    { id: 3, name: 'Charlie', designation: 'Manager' }
  ];
  return (
    <div>
      <h2>Employees List</h2>
      {employees.map(emp => (
        <EmployeeCard key={emp.id} employee={emp} />
      ))}
    </div>
  );
}
export default EmployeesList;
```

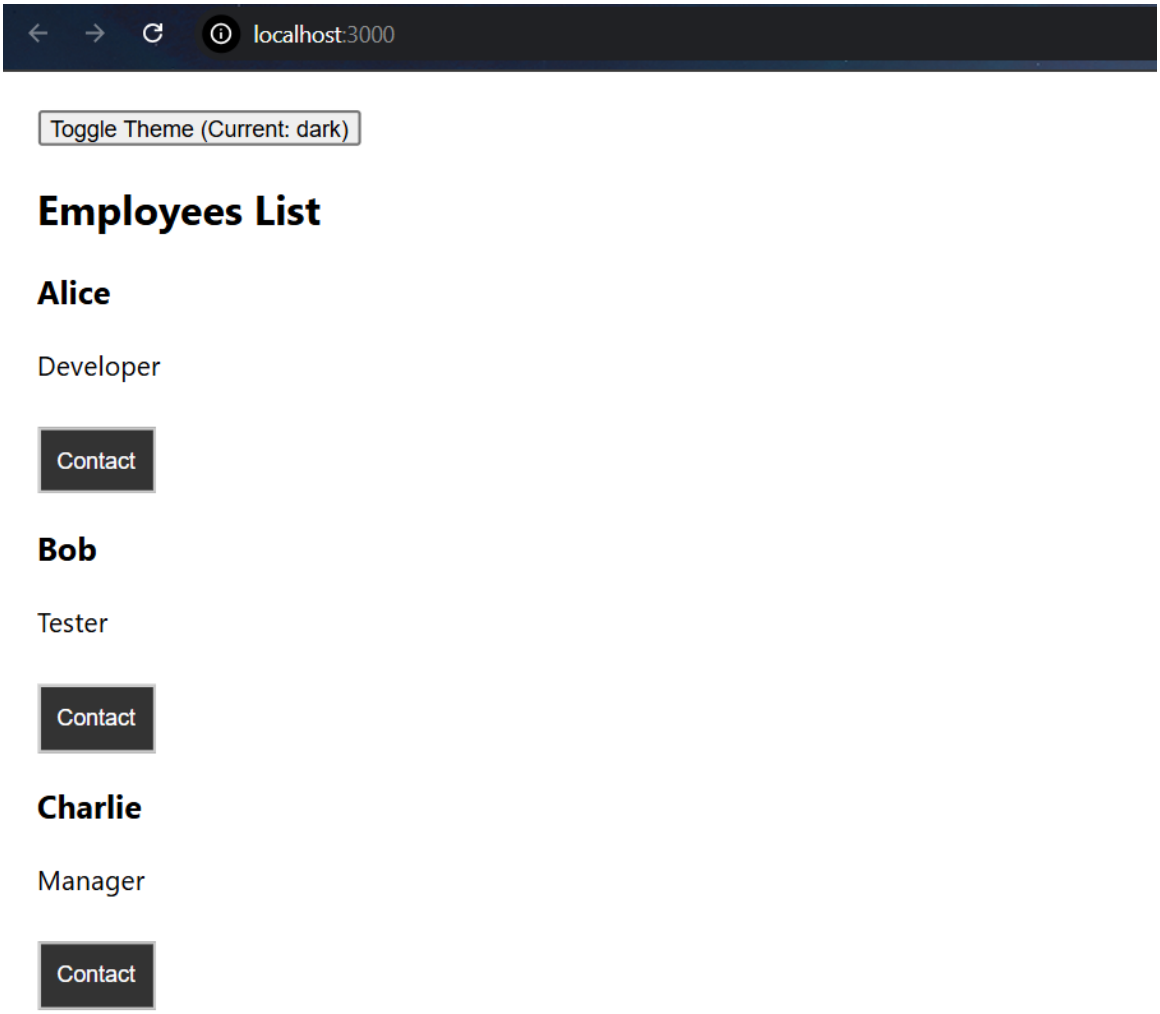
➔ThemeContext.js

```
// src/ThemeContext.js
import React from 'react';
const ThemeContext = React.createContext('light');
export default ThemeContext;
```

Output


ThemeContext.js U
App.js M
EmployeesList.js U
EmployeeCard.js U
contextapp > src > App.js > App
// src/App.js
import React, { useState } from 'react';
import EmployeesList from './EmployeesList';
import ThemeContext from './ThemeContext';
function App() {
const [theme, setTheme] = useState('light');
const toggleTheme = () => {
setTheme(prev => prev === 'light' ? 'dark' : 'light');
};
return (
<ThemeContext.Provider value={theme}>
<div style={{ padding: "20px" }}>
<button onClick={toggleTheme}>
Toggle Theme (Current: {theme})
</button>
<EmployeesList />
</div>
</ThemeContext.Provider>
);
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
node - contextapp
You can now view contextapp in the browser.
Local: http://localhost:3000
On Your Network: http://172.22.32.1:3000
Note that the development build is not optimized.
To create a production build, use npm run build.
webpack compiled successfully
You have Docker installed
install the recommended e
Not Committed Yet Ln 19, Col 26 Spaces: 2 UTF-8 LF JavaScript Go Live


localhost:3000
Toggle Theme (Current: light)
Employees List
Alice
Developer
Contact
Bob
Tester
Contact
Charlie
Manager
Contact


localhost:3000
Toggle Theme (Current: dark)
Employees List
Alice
Developer
Contact
Bob
Tester
Contact
Charlie
Manager
Contact